A Lightweight Hybrid Convolutional Neural Network for Hyperspectral Image Classification

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***Abstract*— Recent studies have demonstrated the potential of hybrid convolutional models that combine 3-D and 2-D convolutional neural networks (CNNs) for hyperspectral image (HSI) classification. However, these models do not fully utilize the benefits of hybrid convolution due to inefficient connections between the two types of CNNs. Moreover, most CNNs, including hybrid models, require a significant number of parameters and computational resources for accurate classification, which increases the need for labeled samples and computational cost. Although the common lightweight strategies, such as depthwise separable convolution (DSC), can reduce parameters and computation compared with normal convolution (NC), they often compromise accuracy. To address these challenges, we propose a lightweight hybrid convolutional neural network (Lite-HCNet) for HSI classification with minimal model parameters and computational effort. First, we design a novel channel attention module (NCAM) and combine it with a convolutional kernel decomposition (CKD) strategy to propose a lightweight and efficient DSC (LE-DSC) deployed in Lite-HCNet. The LE-DSC not only reduces the DSC volume further, but also enhances its performance. Second, a lightweight and efficient hybrid convolutional layer (LE-HCL) is designed in Lite-HCNet to explore the efficient connection structure between 3-D CNNs and 2-D CNNs. Experiments show that the Lite-HCNet reduces the required computational cost and practical deployment difficulty while offering advanced performance with a small number of training samples. Furthermore, abundant ablation experiments confirm the superior performance of the designed LE-DSC.**

***Index Terms*— Hyperspectral image (HSI) classification, lightweight hybrid convolutional neural network (Lite-HCNet), optimized DSC, small training samples.**

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I. Introduction

THE numerous spectral bands of hyperspectral imagery (HSI) bring a wealth of spectral information, greatly improving the ability to extract feature information. Therefore, HSI is favored in various practical applications, including crop analysis [1], soil salinity estimation [2], greening detection [3], vegetation monitoring [4], and urban development [5]. Among the many potential scenarios for HSI applications, research on HSI classification has become increasingly active in recent years.

In early works on HSI classification, researchers mainly explored HSI classification using its rich spectral information, such as support vector machines (SVMs) [6] and random forests [7]. However, the large number of HSI spectral bands and the correlation between spectral bands also bring information redundant and increase the complexity in data processing. Therefore, dimensionality reduction methods were used to extract valuable features, such as principal component analysis (PCA) [8] and linear discriminant analysis (LDA) [9]. To obtain good performance, the spatial information of HSI was gradually introduced into HSI classification and further enhanced its performance. However, traditional feature extraction methods often rely on the prior knowledge of experts in related fields [10], which may limit generalization performance.

In recent years, more and more researchers have used deep learning in the field of computer vision [11], [12], including HSI classification [13], [14], [15], [16], [17], due to its powerful feature learning and extraction capabilities. Among various deep learning models, convolutional neural network (CNN)-based HSI classification models have gained extensive attention, owing to their excellent feature extraction capabilities. Hu et al. [18] designed a deep CNN model to classify HSI directly in the spectral domain and obtained better classification performance than SVMs and traditional deep learning methods. Makantasis et al. [19] used 2-D CNN to encode the spatial–spectral information of HSI pixels with full consideration of the spatial features of HSI pixel neighborhoods. They utilized a multilayer perception for classification. However, the 2-D CNN model changes the three dimensions of HSI and does not completely fuse the spatial–spectral information. Therefore, Kanthi et al. [20] constructed a 3-D-deep CNN model for HSI classification, applying 3-D CNN to extract spatial–spectral features, which greatly improved HSI classification accuracy. He et al. [21] proposed a multiscale

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deep 3-D CNN (M3D-CNN) to further improve the HSI classification performance. The M3D-CNN jointly learns multiscale 2-D spatial and 1-D spectral features from HSI in an end-to-end manner. In addition, Zhong et al. [22] proposed a spatial–spectral residual network (SSRN) that learns to extract deep spatial–spectral features of HSI through continuous spatial–spectral residual blocks. Other CNN models, such as [23] and [24], have also shown superior performance in HSI classification task. However, the aforementioned models only utilized a single CNN. To effectively combine different types of CNNs, Roy et al. [10] proposed a hybrid convolution network (HybridSN), which combines 3-D CNN and 2-D CNN to jointly extract spatial–spectral features and greatly enhance the classification accuracy. However, in HybridSN, 3-D CNN and 2-D CNN are simply stacked and lack efficient connectivity to take full advantage of feature extraction by hybrid convolution.

It is worth noting that CNN models, including hybrid convolutional models, have hundreds of thousands or even millions, of parameters. Plentiful parameters not only require more memory, but also an increase in the number of training samples. Therefore, reducing model size while maintaining performance is a priority. Recent studies have addressed this issues. For example, Ahmad et al. [25] developed a fast and compact 3-D CNN model that reduces parameters and convergence time without sacrificing performance. Jia et al. [26] proposed a lightweight CNN (LWCNN) for HSI classification. They used the spatial–spectral Schroedinger eigenmaps (SSSEs) and dual-scale convolution modules to reduce the number of model parameters while acquiring features, leading to higher efficiency and robustness. Meng et al. [27] proposed a lightweight spectral–spatial convolution module (LS^{2}CM) that serves as an alternative to the convolutional layer. The proposed LS^{2}CM can greatly reduce network parameters and computational complexity in terms of multiply accumulate operations (MACs) while still maintaining or even improving the classification performance. Cui et al. [28] built a lightweight spectral–spatial attention network (LSSAN) based on MobileNetV3 that reduces computational effort and parameters while preserving excellent accuracy. Wang et al. [29] proposed a hybrid convolutional model using single-layer 3-D CNN and 2-D CNN to decrease the model complexity, designed SS-GloRe-Unit to capture global features, and achieved excellent classification results.

Depthwise separable convolution (DSC) is also used as a common lightweighting strategy for the related models. Ghaderizadeh et al. [30] nicely reduced the complexity of the hybrid convolution model by introducing DSC and fast convolution blocks, while using a series of optimization methods to alleviate overfitting and improve classification accuracy. Cui et al. [31] constructed a LiteDepthwiseNet based on 3-D DSC, decomposed standard convolution into depth convolution and point-by-point convolution to obtain high classification performance with minimal parameters, and introduced focal loss to enhance the model's focus on difficult samples and unbalanced data. Although DSC can significantly reduce model volume, this often comes at the expense of accuracy. To overcome this, the above models all use a combination of DSC and normal convolution (NC) to avoid accuracy degradation, but this can increase model volume. Therefore, it is a matter of interest to optimize DSC to have higher performance while keeping it lightweight.

To address the above challenging problems, we propose a lightweight hybrid convolutional neural network (Lite-HCNet) for HSI classification. Our contributions can be summarized as follows.

1) We have optimized the DSC to design a lightweight and efficient DSC (LE-DSC). We employ the convolutional kernel decomposition (CKD) strategy to obtain finer multiscale feature information while reducing the parameters and computational effort. Meanwhile, we designed a new channel attention module tailored to the characteristics of HSI in CNN. By combining the novel channel attention module (NCAM) with DSC, the interaction between spatial and spectral information of HSI was enhanced, resulting in a significant improvement in feature extraction capability.
2) A lightweight and efficient hybrid convolutional layer (LE-HCL) based on LE-DSC is designed to effectively connect 3-D CNN and 2-D CNN for fully utilizing the feature extraction advantages of hybrid convolution. In the LE-HCL, a single-layer 3-D CNN and 2-D CNN are used to jointly extract the spatial spectral features of HSI and filter redundant features by dimensional expansion and compression. Finally, the optimization problem is mitigated by connecting branches through residuals.
3) We constructed Lite-HCNet for HSI classification based on the multiscale parallel LE-HCL. All the convolutions contained in the Lite-HCNet are built using the LE-DSC, which contains only a small number of model parameters and computational effort. This greatly reduces the required computational resources and the practical deployment difficulty. The feature extraction network with LE-HCL as the core ensures excellent classification performance of the model, especially under a small number of training samples.

The rest of this manuscript is organized as follows. In Section II, we briefly describe related works. In Section III, we provide a detailed description of Lite-HCNet, LE-HCL, and LE-DSC. Section IV reports relevant experimental results and carries out corresponding analysis and discussion. Finally, we summarize the manuscript in Section V.

II. Relative Work

To better understand the work we have done, in this section, we briefly introduce some DSC and channel attention concepts and along with their relevant literature.

A. Depthwise Separable Convolution

Convolution is a widely used and effective method for feature extraction in various applications, such as target detection [32], image classification [33], image segmentation [34], and remote sensing image processing [35]. Fig. 1 shows the simple operation of NC, including 2-D and 3-D convolution. The 3-D convolution operates on the same principle as 2-D convolution, with the difference that both the input and output data and the convolution kernel are in 3-D. However, for the model

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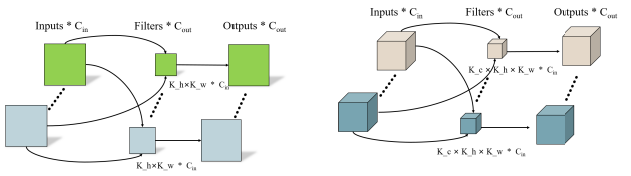


Fig. 1. NC operation. (a) 2-D NC. (b) 3-D NC.

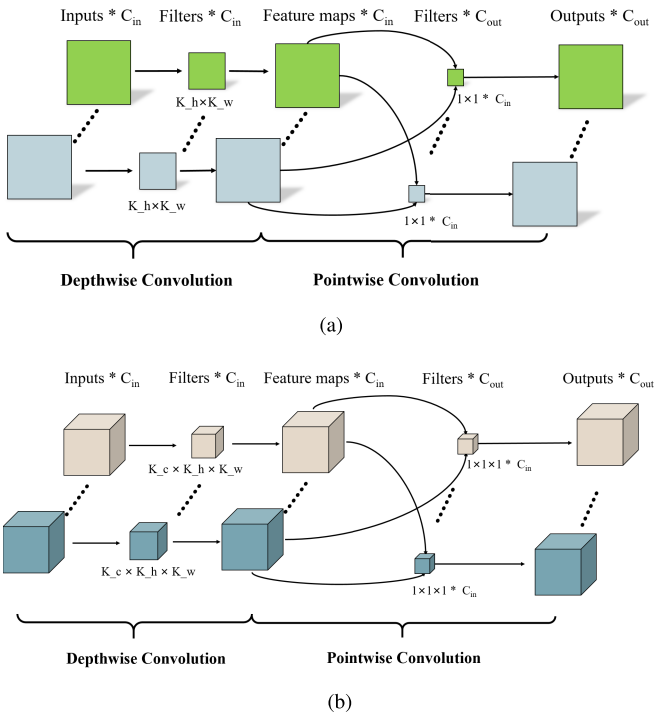


Fig. 2. DSC operation. (a) 2-D DSC. (b) 3-D DSC.

lightweighting, a new convolution operation called DSC was proposed in [36].

Unlike NC, DSC decomposes the convolution operation into two parts: depthwise convolution and pointwise convolution. Fig. 2 shows the schematic of the simple operation of DSC, including 2-D and 3-D convolution, respectively. Specifically, assuming that the input data have C_{in} channels, each channel in depthwise convolution will be treated as a separate input with a convolution kernel for convolution and output. In pointwise convolution, all channels are convolved with a convolution kernel size of 1 and output a result, so that there are C_{out} of convolution kernels; i.e., the final number of channels is C_{out}. This operation of DSC greatly reduces the number of parameters and computation compared with NC, so it is widely used in model lightweight design, such as [30] and [31], and the related structure diagram is shown in Fig. 3.

Although DSC greatly reduces the parameters and computational effort, it has some limitations. For instance, in deep convolution, the connections between channels are isolated, while in point convolution, the information of spatial regions between different channels is not sufficiently connected. As a result, DSC can lead to performance degradation while reducing the volume. Based on this, the two models shown in Fig. 3 contain NC in their structures besides DSC to balance the volume and performance of the models.

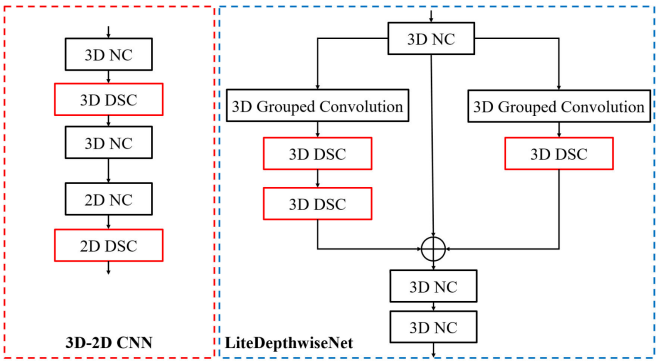


Fig. 3. Related model structure.

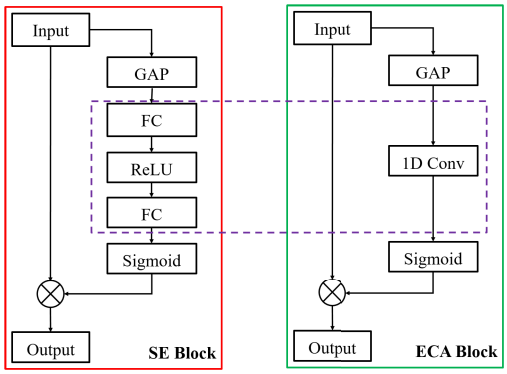


Fig. 4. Related channel attention structure.

B. Channel Attention

The attention module has been widely adopted in various network models, achieving promising results. Essentially, an attention module is a set of weights that enhances effective features and weakens ineffective ones. Therefore, a properly designed attention module can significantly improve the performance of a model. The attention module can alleviate the issue of insufficient feature association that exists in DSC. They are typically divided into two categories: spatial attention and channel attention. Combining the complex spectral channel information of HSI and the sensitive and hard-to-learn characteristics of spatial attention, we focus solely the channel attention module in this discussion. Fig. 4 illustrates the structure diagrams of the two classical channel attention modules: squeeze-and-excitation (SE) [37] and efficient channel attention (ECA) [38].

From Fig. 4, we can observe that the channel attention process involves three primary steps. First, the input data are abstracted from the overall features to a series of point features using global average pooling (GAP). Then, the connection between these point features is established by operations, and finally, attention weights are obtained and attached to the input data by normalizing them with the sigmoid function. The second step is key, as it determines how to establish the connection between these point features through operations. As shown in Fig. 4, the difference between the SE block and the ECA block is in the purple dashed box, which is the second step of the operation. SE block is a global operation to obtain the association between point features by fully connected (FC), while the ECA block is a local operation by 1-D convolution (Conv). The FC operation is performed on all the features, which has the advantage of obtaining global information, but

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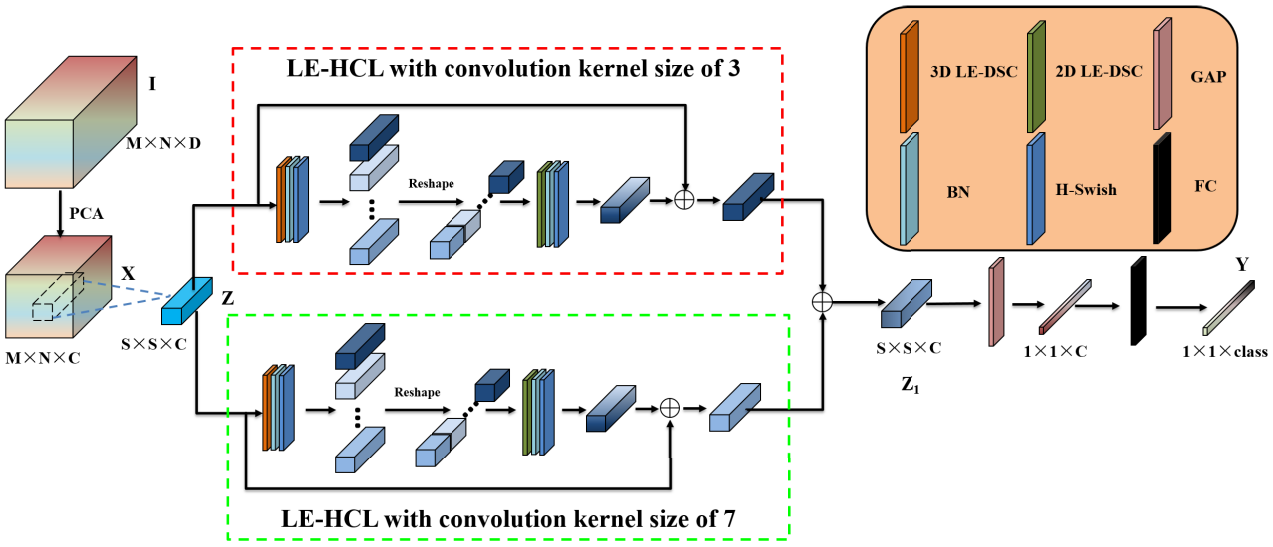


Fig. 5. Overall structure diagram of the Lite-HCNet for HSI Classification. Lite-HCNet based on two LE-HCLs in parallel at different scales is designed to jointly extract multiscale HSI spatial–spectral features. Lite-HCNet reduces the number of parameters, alleviates practical deployment difficulties, and achieves satisfactory performance, significantly.

this does not avoid an increase in the number of operations. In contrast, the 1-D Conv operation is performed locally on the features, and although the amount of operations is reduced, some information is also missing. It follows that a suitable operation for establishing the connection between point features urgently needs to appear.

III. Methods

In this section, we will first present a general overview of the proposed Lite-HCNet, followed by the detailed structures of LE-HCL and LE-DSC are given, respectively.

A. Lightweight Hybrid Convolutional Neural Network

In this manuscript, we designed a Lite-HCNet for HSI classification, which is both simple and efficient. All convolutions included in Lite-HCNet use the LE-DSC we designed to reduce the volume of the model. In addition, the feature extraction network with LE-HCL structure as the core ensures the performance of the Lite-HCNet. Fig. 5 shows a detailed illustration of the proposed Lite-HCNet.

Let the original HSI be denoted as $\boldsymbol{I} \in R^{M\times N\times D}$, where M and N are the width and height of spatial dimensions, and D is the number of spectral bands (generally in the tens or hundreds). Although the many spectral bands of HSI bring abundant information, the excessive number also generates information redundancy and produces high-computational costs for the subsequent processing task. Moreover, when the training samples are limited, the overall classification accuracy of HSI tends to decrease, as the feature dimensions increase, i.e., the "dimensional disaster" problem. Therefore, it is necessary to reduce information redundancy by decreasing HSI's spectral dimension. The core idea of PCA, a common dimension reduction method, is to extract the key features by calculating the similarity between different data. Therefore, in Lite-HCNet, we use PCA as the first step to reduce the dimensions of HSI's spectrum and extract the key features. This effectively solves the problems of high-computational cost and low classification accuracy caused by information redundancy. After PCA processing, the HSI is represented as $\boldsymbol{X} \in R^{M\times N\times C}$. The spatial dimension remains unchanged, and the number of spectral bands is reduced from D to C.

To better utilize the subsequent CNN for classification, the HSI is preprocessed with neighborhood block extraction after PCA. That is, for each pixel of HSI, a patch is extracted with its center point as the input data of the subsequent CNN. The label of each patch is the label of the center pixel, denoted as $\boldsymbol{Z} \in R^{S\times S\times C}$. The width and height of the patch are S, and the number of spectral bands is C.

In the feature extraction network, we implemented an LE-HCL based on a rhombic structure (extended–compression) to improve the connection efficiency between the 3-D CNN and the 2-D CNN. In the LE-HCL, an LE-DSC of our design is used to convolve and extract the spatial–spectral features of the HSI. The complete feature extraction network structure is shown in Fig. 5 and is a multiscale parallel structure built on the basis of the designed LE-HCL. By setting up convolution kernels of different sizes in the LE-HCL, the HSI spatial–spectral features are fully extracted from multiple scales. Specifically, we construct two scales of LE-HCL in Lite-HCNet, as shown in the red and green dashed boxes in Fig. 5. Specifically, as shown in Fig. 5, we build two LE-HCLs of two scales in Lite-HCNet. The convolution kernels of the two LE-HCLs are 3 and 7, respectively [taking the LE-HCL with a convolution kernel size of 3 as an example; i.e., the convolution kernel size in 3-D LE-DSC is $(3 \times 3 \times 3)$ and in 2-D LE-DSC is (3×3)]. The LE-HCL with a convolutional kernel size of 3 is used to extract a small range of local feature information, while LE-HCL with a convolutional kernel size of 7 is used to extract a larger range of feature information. The two are fused to output a more information-rich feature map, and the output $\boldsymbol{Z_1} \in R^{S\times S\times C}$ can be denoted as follows:

$$\boldsymbol{Z_1} = f^{3}_{\text{LE_HCL}}(\boldsymbol{Z}) + f^{7}_{\text{LE_HCL}}(\boldsymbol{Z}) \tag{1}$$

where $f^{3}_{\text{LE_HCL}}(\cdot)$ denotes an LE-HCL with a convolution kernel size of 3 and $f^{7}_{\text{LE_HCL}}(\cdot)$ represents an LE-HCL with a convolution kernel size of 7.

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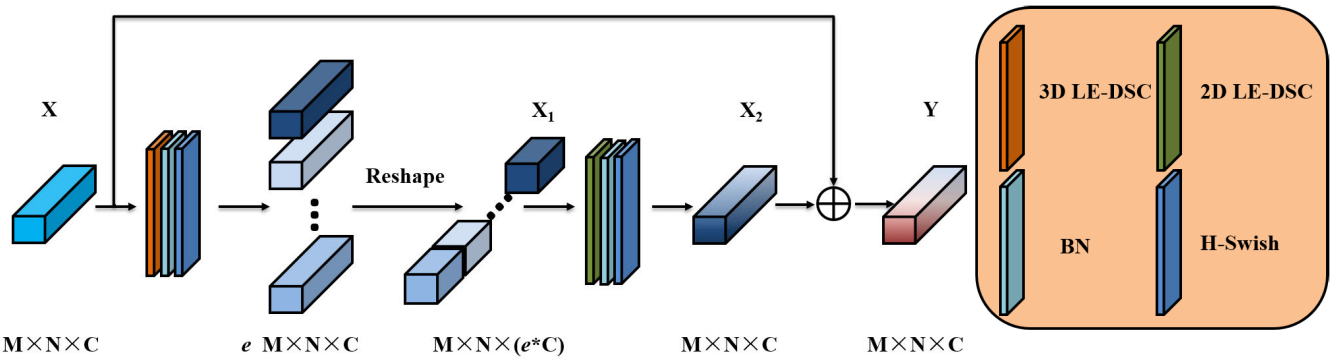


Fig. 6. Structure diagram of LE-HCL. LE-HCL extracts the spatial–spectral features of HSI by single-layer 3-D CNN and 2-D CNN together and captures the high-dimensional spectral features adequately by an extension–compression strategy. Residual connectivity strategy is introduced to improve the feature extraction capability of the module and alleviate the optimization problem.

The classification is performed by the GAP layer and the output of the FC layer. The number of nodes in the FC layer output is the number of categories in the dataset. The final output $\boldsymbol{Y}$ can be expressed as follows:

$$\boldsymbol{Y} = f_{\mathrm{FC}}(f_{\mathrm{GAP}}(\mathbf{Z_1})) \tag{2}$$

where $f_{\mathrm{FC}}(\cdot)$ presents the FC operation and $f_{\mathrm{GAP}}(\cdot)$ indicates the GAP operation.

B. Lightweight and Efficient Hybrid Convolution Layer

The existing hybrid convolutional models that stack 3-D CNN and 2-D CNN suffer from inefficient connection. To address this issue, we propose an LE-HCL structure. The LE-HCL is designed to explore efficient connections of hybrid convolution and take full advantage of its feature extraction capabilities, as shown in Fig. 6. The structure is designed in a diamond-shaped manner, with the wide middle and narrow sides. First, a single layer of 3-D CNN is designed. In the 3-D CNN layer, we do not use 3-D NC but use our designed 3-D LE-DSC to jointly extract spatial spectral features and extend the dimensionality of the input data to enrich the feature information. Second, a single-layer 2-D CNN is designed. Instead of using 2-D NC, we use the designed 2-D LE-DSC to further extract spatial features and compress the dimensionality back to the original dimension for filtering redundant features. Finally, a residual connection branch is designed to alleviate the optimization problem. Suppose the input HSI data $\boldsymbol{X} \in R^{S\times S\times C}$, where $S \times S$ is the spatial dimension, i.e., width $\times$ height, and C denotes the spectral dimension; the specific design details of the LE-HCL are as follows.

1) To ensure the integrity of the HSI 3-D data features, we first design a layer of 3-D CNN. The spatial–spectral features of HSI are extracted jointly by the designed 3-D LE-DSC, and the spectral dimension of the input data is extended and mapped to a higher dimensional space to obtain more adequate spectral feature with an extension factor of e. The H-swish [39] activation function is then applied to introduce nonlinearity and enhance feature representation, with batch normalization introduced before the activation function to accelerate network convergence. As shown in Fig. 6, the dimensionally expanded data are represented as $\boldsymbol{X_1} \in R^{S\times S\times(e\times C)}$.

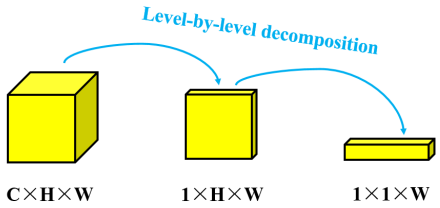


Fig. 7. Level-by-level decomposition strategy.

2) To further extract the spatial features of HSI, we also designed a layer of 2-D CNN. The 2-D LE-DSC was used to extract the features from the dimensionally expanded data and to improve the spatial feature representation. The spectral dimension of the input data is compressed back to the original dimension, and the output is $\boldsymbol{X_2} \in R^{S\times S\times C}$. After that, the H-swish activation function and batch normalization are applied. This rhombic structure not only can filter the redundant information and enhances feature extraction, but also can reduce the number of relevant parameters and computational effort, ensuring a lightweight structure.
3) To make the module more efficient, a residual connection branch was set up, connecting the raw input and the output to alleviate the optimization problem.

The LE-HCL aggregates single-layer 3-D CNNs and 2-D CNNs and provides a more efficient way to connect them. As a result, LE-HCL can be used as a novel "module" on which to build more complex and efficient hybrid convolutional models quickly. The output $\boldsymbol{Y}$ after LE-HCL can be expressed as follows:

$$\boldsymbol{Y} = \boldsymbol{X} + \sigma(f_{\mathrm{BN}}(f_{2-\mathrm{D}}(R(\sigma(f_{\mathrm{BN}}(f_{3-\mathrm{D}}(\boldsymbol{X})))))) \tag{3}$$

where $\sigma(\cdot)$ represents the H-swish activation function, $f_{\mathrm{BN}}(\cdot)$ indicates the batch normalization, $R(\cdot)$ denotes the dimensional transformation function, $f_{2-\mathrm{D}}(\cdot)$ indicates the 2-D LE-DSC function, and $f_{3-\mathrm{D}}(\cdot)$ is the 3-D LE-DSC function.

C. Lightweight and Efficient Depthwise Separable Convolution

As mentioned in Section II, although DSC reduces the parameters and computation compared with NC, its structure leads to insufficient isolation and interaction between feature information, resulting in some performance degradation.

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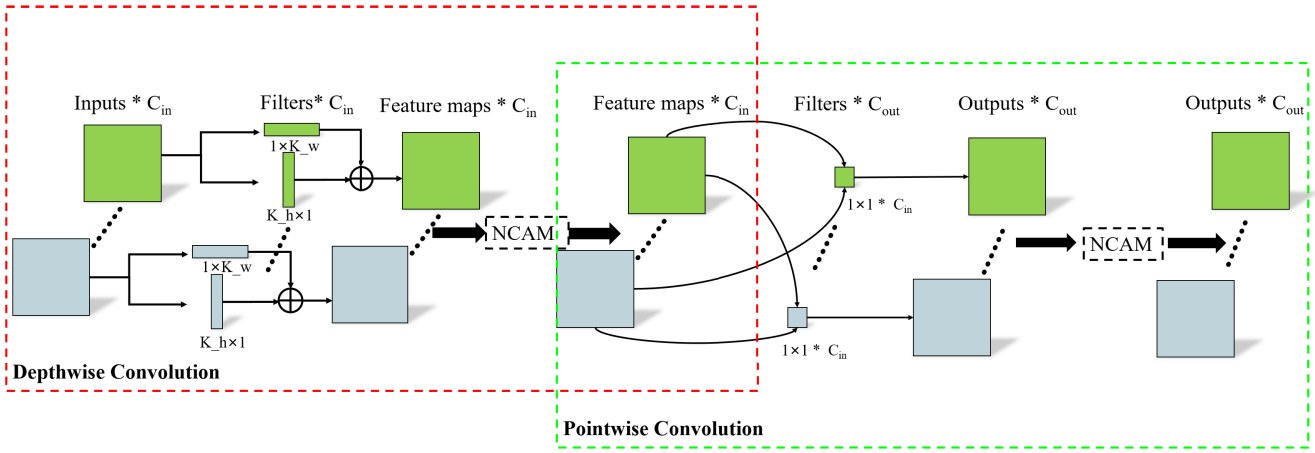


Fig. 8. Operational structure diagram of 2-D LE-DSC. CKD strategy is used to obtain finer multiscale feature information while reducing parameters and computational effort. NCAM is integrated into the depthwise and pointwise convolution, respectively, to enhance the interaction between spatial and spectral feature information for improving feature extraction capability.

To address this, we optimized DSC by first decompose the convolutional kernels in the depthwise convolution into parallel multiple convolutional kernels, which obtains finer multiscale feature information while further reducing the parameters and computational effort. It is worth noting that our strategy for CKD follows the principle of level-by-level decomposition (as shown in Fig. 7), which ensures that good performance is preserved. The depth convolution part of DSC suffers from a problem of isolating spectral information between channels, while the point convolution part faces a problem of isolating spatial information between channels. To address this issue, we design an NCAM that fuses the HSI features in CNN and integrate it into the deep and point convolutions of DSC. The NCAM enables more comprehensive spectral as well as spatial information interaction between different channels, thus significantly improving the feature extraction capability. The detailed structural design is as follows.

1) 2-D LE-DSC: Fig. 8 shows the operational structure diagram of 2-D LE-DSC designed for 2-D DSC. Suppose the size of 2-D convolution kernel is $K_h \times K_w$; i.e., the height is K_h, and the width is K_w.

In the depthwise convolution part, we decompose it into two parallel strip convolutions in spatial width and height, respectively, and then add and fuse the results, whose sizes are $1 \times K_w$ and $K_h \times 1$. This design of decomposing convolutional kernels can further reduce the parameters and computation, which is more and more obvious the larger the convolutional kernel is. At the same time, the decomposed multiple parallel convolutional kernels can extract more detailed features at different scales and obtain richer feature information after fusion. To address the problem that the channel-by-channel convolution in the depthwise convolution part of DSC leads to segregation of channel information, we design NCAM to obtain the weight distribution between different channels after inserting into the depthwise convolution. The weight distribution is appended to the original features to strengthen the interaction between spectral channel feature information.

In the pointwise convolution part, although different channels can be linked with each other by a 1- × 1-sized convolution kernel, the 1- × 1-sized convolution kernel cannot obtain the region information in spatial dimension, which generates a lack of interaction of spatial information between different channels. In contrast, NCAM is a GAP operation to obtain the condensed point feature distribution, and the final generated channel weights contain spatial information. Therefore, we also insert the designed NCAM after pointwise convolution to enhance the interaction of spatial feature information.

2) 3-D LE-DSC: Fig. 9 shows the operational structure of the 3-D LE-DSC designed for the 3-D DSC. The size of the 3-D convolution kernel is assumed to be $K_c \times K_h \times K_w$; i.e., the depth is K_c, the height is K_h, and the width is K_w. The designed 3-D LE-DSC has the same principle as the 2-D LE-DSC, and there are only two differences in the structure design. Therefore, we are not going to elaborate other than that.

First, in the depthwise convolution part of 3-D LE-DSC, the 3-D convolution kernel is decomposed into three faceted convolutions in spatial and spectral dimensions, respectively, and the results are subsequently summed and fused with the sizes of $K_c \times 1 \times K_w$, $K_c \times K_h \times 1$, and $1 \times K_h \times K_w$, respectively.

Second, the position of inserting NCAM is the same as in 2-D LE-DSC, but it is necessary to dimensionally transform the 3-D data to generate 2-D data before its input, in order to obtain more sufficient feature information by computing all spectral channels, and then finally transform it back to the original 3-D dimension. That is, suppose the size of 3-D data before NCAM input is $N \times C \times H \times W$, where N is the number of channels of 3-D data, C is the depth of 3-D data, and H and W are the width and height of 3-D data, respectively. Then, the size of the data after dimensional transformation is $(N \times C) \times H \times W$, where $(N \times C)$ is the number of channels, and H and W are the width and height, respectively.

3) NCAM: Channel attention can effectively mitigate the issue of inadequate feature information interaction within DSC, but the existing channel attention module lacks a suitable algorithm to establish connections between channel point features. Therefore, we design an NCAM to address this problem. NCAM processes data in 2-D. NCAM considers the characteristics of HSI in CNN in its structure design. We design two branches, global branch and local branch, to process all input data and partial input data, respectively, and fuse them to generate a more appropriate distribution

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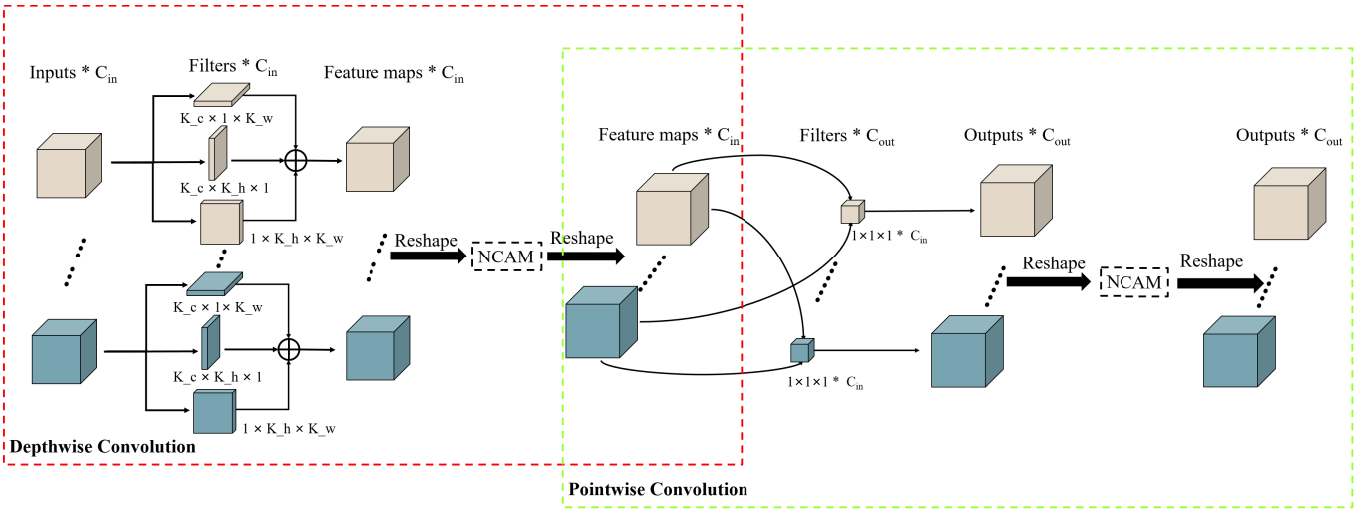


Fig. 9. Operational structure diagram of 3-D LE-DSC. CKD strategy is used to obtain finer multiscale feature information while reducing parameters and computational effort. NCAM is integrated into the depthwise and pointwise convolution, respectively, to enhance the interaction between spatial and spectral feature information for improving feature extraction capability.

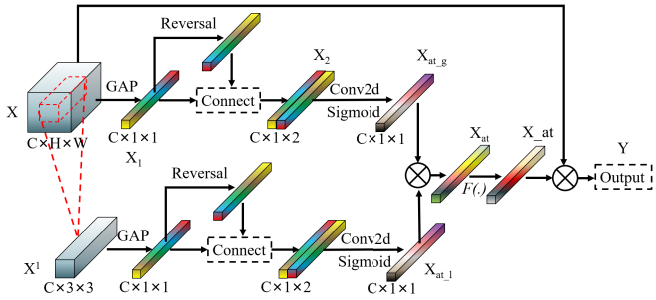


Fig. 10. Operational structure diagram of NCAM. Global and local channel attentions are generated in two branches, which are fused to achieve more appropriate channel attention. Head-to-tail reversal splicing is used to capture both local channel information and longer range channel information.

of channel attention weights. Fig. 10 shows the operational structure diagram of NCAM. Let the 2-D input be $\boldsymbol{X} \in R^{C\times H\times W}$ of NCAM, where C is the number of channels, and H and W stand for height and width, respectively.

In the global branch, i.e., the upper branch in Fig. 10, we obtained the characteristic distribution of concentrated channel points of $\boldsymbol{X}$ by using GAP, denoted as $\boldsymbol{X_1} \in R^{C\times 1\times 1}$. In [37], the FC uses all channels as input. This is certainly the most comprehensive in terms of information, but this inevitably introduces more parameters and potential redundancy. In [38], 1-D Conv is used for local operations, and the size of the convolution kernel determines the size of the range from which information is extracted. This reduces the parameters but also reduces the amount of information colluded from more distant ranges compared with FC. For example, in order to obtain the association of head and tail channel features, a convolution kernel including the whole channel is required, which is not efficient. In contrast to these two approaches, we propose a head and tail reverse splicing structure, which also employs a local convolution kernel strategy. However, this novel structure can not only extract the local channel information around its own node, but also obtain the channel information in a more distant range. It is a more efficient structure. Specifically, there are two steps.

Step 1: We obtain $\boldsymbol{X_2} \in R^{C\times 1\times 2}$ by inverting the head and tail of $\boldsymbol{X_1}$ and splicing it with itself.

Step 2: We design a 2-D convolution kernel $\boldsymbol{K} \in R^{2\times K_w}$ to convolve it and obtain the normalized global attention weight $\boldsymbol{X_{at_g}} \in R^{C\times 1\times 1}$ by using the sigmoid function. In fact, this is a 1-D convolution using a 2-D convolution kernel, which can be thought of as a 1-D convolution kernel of size $2 \times K_w$. The size of K_w is chosen adaptively based on the channel length of the input data, and the longer the channel, the larger the K_w is, denoted as follows:

$$K_w = \left| \frac{(\log_2 C + b)}{\gamma} \right|_{\text{odd}} \tag{4}$$

where C is the number of channels and the $|\cdot|_{\text{odd}}$ function is used to ensure that K_w is an odd number. In this article, b and γ are set to 2 and 3, respectively.

The typical attention module only contains the global branches mentioned above. However, because of some special characteristics of HSI classification in CNN, the typical attention module is not exactly suitable. During processing, HSI is usually segmented into neighborhood blocks, where the label of the central pixel point of the neighborhood block represents the label of the entire block. Therefore, it is critical to extract channel attention from the central pixel and its spectral information. To achieve this, we design a local branch to obtain the spectral channel attention weight distribution of the central pixel. However, extracting only the center pixel may introduce some instability. To increase the fault tolerance, we assume that neighboring pixels are more likely to have the same label, and consider the center pixel and its local neighborhood information as a replacement for the center pixel. In this article, we set this neighborhood size to 3×3 and extract the center of the input $\boldsymbol{X}$ and its surrounding neighborhood blocks of size 3×3 as the input of the local branch, denoted as $\boldsymbol{X^1} \in R^{C\times 3\times 3}$. The subsequent operations are the same as those of the global branch. The local attention weights $\boldsymbol{X_{at_l}} \in R^{C\times 1\times 1}$ are achieved.

The local and global attention weights are multiplied to generate the bootstrap-enhanced attention weights $\boldsymbol{X_{at}} \in R^{C\times 1\times 1}$, which are then passed through the $F(\cdot)$ function to increase the gap between the attention weights and highlight the more important channel features. The resulting fused attention

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TABLE I
CATEGORY AND QUANTITY ON IP, UP, HU, AND YRD DATASETS

No.	Indian Pines Classes	Samples	No.	University of Pavia Classes	Samples	No.	Houston 2013 Classes	Samples	No.	Yellow River Delta Classes	Samples
1	Alfalfa	46	1	Asphalt	6631	1	Healthy grass	1251	1	Spartina alterniflora	46998
2	Corn-notill	1428	2	Meadows	18649	2	Stressed grass	1254	2	Water body	255986
3	Corn-min	830	3	Gravel	2099	3	Synthetic grass	697	3	Reed swamp	48024
4	Corn	237	4	Trees	3064	4	Trees	1244	4	Suaeda salsa	34923
5	Grass/Pasture	483	5	Painted Metal sheets	1345	5	Soil	1242	5	Saline alkali Beach	36051
6	Grass/Trees	730	6	Bare Soil	5029	6	Water	325	6	Bare tidal flat	94318
7	Grass/pasture-mowed	28	7	Bitumen	1330	7	Residential	1268	7	Sparse tidal flat	15740
8	Hay-windrowed	478	8	Self-Blocking Bricks	3682	8	Commercial	1244	8	Tamarix	37760
9	Oats	20	9	Shadows	947	9	Road	1252			
10	Soybeans-notill	972				10	Highway	1227			
11	Soybeans-min	2455				11	Railway	1235			
12	Soybeans-clean	593				12	Parking Lot 1	1233			
13	Wheat	205				13	Parking Lot 2	469			
14	Woods	1265				14	Tennis Court	428			
15	Bldg-Grass-Tree-Drives	386				15	Running Track	660			
16	Stone-steel towers	93									
	TOTAL	10249		TOTAL	42776		TOTAL	15029		TOTAL	569800

weights are denoted as $\boldsymbol{X_at} \in R^{C\times 1\times 1}$

$$\boldsymbol{X_at} = F(\boldsymbol{X_{at}}) = \text{Sigmoid}((\boldsymbol{X_{at}} - \alpha) \times \beta). \tag{5}$$

Since the input $\boldsymbol{X_{at}}$ of (5) is distributed between 0 and 1, if it is directly fed into the sigmoid function, the output value will be distributed between 0.5 and 1. This does not widen the gap between the different weights, but rather makes them more aggregated. Therefore, we introduce two additional parameters, α and β. The role of α is to make some values of $\boldsymbol{X_{at}}$ negative, while β can widen the distribution of $\boldsymbol{X_{at}}$. This leads to a larger gap between different weights. For the values of α and β, we take the manual setting, where α is taken to the median of the $\boldsymbol{X_{at}}$ distribution as much as possible and β should not be too large. In this article, α is set to 0.2, and β is set to 2. The generated attention weights $\boldsymbol{X_{at}}$ are appended to the input $\boldsymbol{X}$ and the output $\boldsymbol{Y}$, denoted as follows:

$$\boldsymbol{Y} = \boldsymbol{X_at} \odot \boldsymbol{X} \tag{6}$$

where $\odot$ is the elementwise multiplication.

IV. EXPERIMENTS AND ANALYSIS

This section presents the datasets used in the experiments, the model parameter settings, and focuses on discussing in detail the results compared with various methods on these datasets.

A. Datasets Description

We used three public HSI datasets and one Yellow River Delta (YRD) dataset for our experiments. The three public HSI datasets are Houston 2013 (HU), University of Pavia (UP), and Indian Pines (IP), respectively. The HU dataset has a spatial dimension of 349 × 1905, 144 spectral bands, a wavelength range of 380–1050 nm, and includes 15 feature classes. The UP dataset has a spatial dimension of 610 × 340, 103 spectral, a wavelength range of 430–860 nm, and includes nine feature categories. The IP dataset has a spatial dimension of 145 × 145, 200 spectral bands after removing 24 absorbing spectral bands, a wavelength range of 400–2500 nm, and includes 16 feature classes. The YRD dataset has a spatial dimension of 462 × 617, 150 spectral bands, a wavelength range of 390–1029 nm, and includes eight feature classes.

Table I provides a detailed overview of the categories in each dataset and the number of samples in each category.

B. Network Setting

All experiments were performed using PyTorch on a Lenovo Saver-Y7000P laptop equipped with a GTX 1660ti graphics processor (GPU) and 16-GB RAM. We used the cross-entropy loss function to calculate associated losses and optimized the network using the Adam optimizer with the learning rate set to 0.01. *L2* regularization was added and set to 0.0005. The network was trained for 50 epochs using a mini-batch of size 16. The learning rate decay was set to one-tenth of the original learning rate at the 40th epoch for the IP dataset and 25th epoch for the UP and HU datasets, while the 15th epoch for YRD datasets. The expansion factor e was set to 3 for all LE-HCLs in the proposed Lite-HCNet. It is worth noting that no data augmentation or validation sets were developed for any of the datasets. Moreover, to ensure a fair comparison, all other models involved in the comparison experiments were set according to the parameters in their respective papers, except for the training samples ratio.

C. Results and Discussion

We adopted three widely used evaluation metrics to assess the performance of HSI classification: kappa coefficient (k), overall accuracy (OA), and average accuracy (AA). k reflects the consistency of information between messages, OA demonstrates the proportion of correctly classified samples among the total test samples, and AA represents the average proportion of correctly classified samples in each category.

In the CNN-based HSI classification model, different neighborhood block sizes can significantly affect the model's performance. To set more reasonable parameters on Lite-HCNet, we conducted parameter sensitivity experiments on four experimental datasets for the spatial and spectral dimension size of the extracted HSI neighborhood blocks. The experimental results are shown in Fig. 11. We found that the smaller the spatial and spectral dimensions of the HSI neighborhood blocks, the less feature information is available, resulted in poor performance on all four datasets. As the dimensionality increases, the accuracy on all four datasets increases

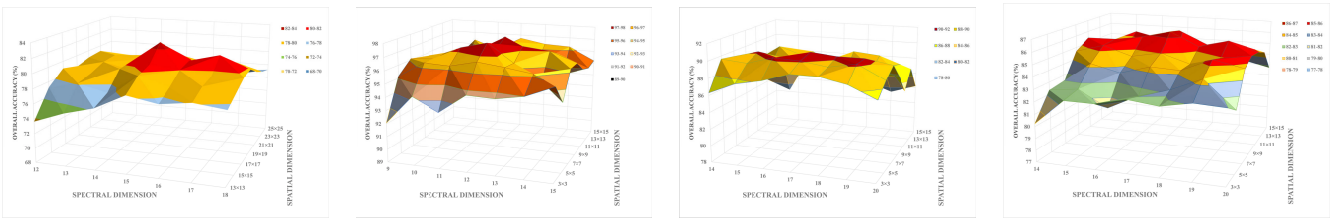
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TABLE II
CLASSIFICATION ACCURACIES (IN PERCENTAGES) ON IP, UP, HU, AND YRD (1% SAMPLES FOR TRAINING)

Dataset		2D CNN	HybridSN	3D-2D CNN	LSSAN	F3D	MSDAN	LiteDepthwiseNet	Lite-HCNet
Indian Pines	$k \times 100$	56.39±0.4	61.98±0.6	61.39±1.1	73.76±2.3	58.82±1.3	**80.62±0.7**	77.09±1.6	80.54±0.9
	OA × 100	62.67±0.3	67.45±0.2	66.79±0.7	74.27±1.5	63.78±0.7	82.50±1.1	80.04±1.4	**82.72±0.8**
	AA × 100	52.06±0.5	58.06±0.6	51.48±0.9	61.91±2.4	60.96±0.9	77.45±1.2	71.82±0.3	**83.46±1.0**
University of Pavia	$k \times 100$	88.71±0.3	92.49±0.4	96.35±0.1	92.65±0.4	93.73±0.2	94.61±0.2	96.29±0.2	**96.86±0.1**
	OA × 100	91.58±0.3	94.58±0.3	96.92±0.2	94.23±0.3	95.44±0.4	96.29±0.2	96.93±0.2	**97.65±0.1**
	AA × 100	87.86±0.5	88.25±0.4	95.16±0.1	92.09±0.4	91.07±0.7	94.95±0.2	94.53±0.4	**96.66±0.2**
Houston 2013	$k \times 100$	77.99±0.4	81.72±0.4	81.54±0.6	82.79±0.4	80.46±0.3	88.97±0.1	83.44±0.4	**90.02±0.3**
	OA × 100	79.60±0.2	82.81±0.2	82.42±0.3	82.00±0.3	81.96±0.3	90.42±0.4	85.18±0.9	**90.51±0.6**
	AA × 100	75.57±0.3	81.05±0.6	80.73±0.5	82.33±0.4	81.15±0.7	90.17±0.2	84.94±0.8	**90.31±0.6**
Yellow River Delta	$k \times 100$	65.50±0.3	80.08±0.4	71.81±0.1	79.38±0.4	80.36±0.3	77.72±0.4	80.30±0.2	**81.70±0.2**
	OA × 100	72.55±0.3	85.11±0.1	78.95±0.2	84.52±0.3	85.02±0.4	83.65±0.3	85.17±0.1	**86.06±0.2**
	AA × 100	58.75±0.3	80.23±0.4	71.74±0.2	79.77±0.6	80.90±0.4	80.56±0.2	82.13±0.3	**82.24±0.3**

TABLE III
MODEL COMPLEXITY AND EFFICIENCY OF DIFFERENT METHODS ON IP, UP, HU, AND YRD (1% SAMPLES FOR TRAINING)

Dataset		2D CNN	HybridSN	3D-2D CNN	LSSAN	F3D	MSDAN	LiteDepthwiseNet	Lite-HCNet
Indian Pines	Params/K	295	5122	1320	199	28520	3424	52	**3**
	Train time/m	8.35	9.17	7.15	16.36	4.31	14.48	10.33	**4.25**
	FLoPs/M	**62**	127053	26738	889	99184	77205	3112	70
University of Pavia	Params/K	33	4844	1043	39	994	3424	30	**2**
	Train time/m	9.08	8.16	8.03	15.52	**4.38**	18.27	12.15	4.56
	FLoPs/M	7	26085	10459	249	5166	41518	1508	**7**
Houston 2013	Params/K	1343	557	556	40	1339	3424	31	**3**
	Train time/m	11.22	9.45	9.49	16.42	5.37	16.29	12.43	**5.29**
	FLoPs/M	1013	5171	7088	258	1106	25570	4696	**11**
Yellow River Delta	Params/K	843	294	293	40	552	3383	17	**3**
	Train time/m	12.20	10.34	9.54	30.37	**5.26**	72.24	14.14	13.42
	FLoPs/M	620	2652	3213	376	3143	38196	835	**27**



(a) (b) (c) (d)

Fig. 11. Effects of spatial and spectral dimensions on Lite-HCNet performance on different datasets. (a) IP. (b) UP. (c) HU. (d) YRD.

to different degrees, indicating the need for the model to have more valid information. However, as the dimensionality increases excessively, the accuracy starts to decrease due to the introduction of more noise, which interfered with the learning of the model. Therefore, according to the results in Fig. 11, we set an appropriate neighborhood block size for Lite-HCNet on each of the four datasets. The width, height, and depth of the neighboring blocks are set to 19, 19, and 15, respectively, on the IP dataset, i.e., 19 × 19 × 15. For the other datasets, UP is set to 7 × 7 × 12, HU is set to 7 × 7 × 17, and YRD is set to 11 × 11 × 17.

To evaluate the performance of Lite-HCNet for HSI classification, we compared it with several popular and state-of-the-art models, including classical hybrid convolutional models, models designed for lightness and speed, and complex models. Specifically, we compared our approach with the following model: 2-D CNN (IGARSS2015) [19], hybrid spectral CNN (HybirdSN) (GRSL2020) [10], 3-D–2-D CNN (JSTARS2021) [30], LSSAN (TGRS2022) [28], fast and compact 3-D CNN (F3D) (GRSL2022) [25], multiscale densely connected attention network (MSDAN) (JSTARS2022) [40], and LiteDepthwiseNet (TGRS2022) [31]. The performance of a deep learning model is heavily dependent on the number of training samples available. A sufficient number of training samples can significantly improve model performance, whereas an insufficient number often results in poor performance. However, manually labeling samples is expensive, which limits the number of training samples available. Therefore, it is essential to compare different models in terms of performance under a small number of training samples. In our study, we compared the performance of various methods using only 1% training samples in different datasets. We conducted experiments to compare $k \times 100$, OA × 100, and AA × 100, and the results are shown in Table II. Table III compares the complexity and efficiency of different models, respectively. It is worth noting that due to the extremely unbalanced distribution of samples

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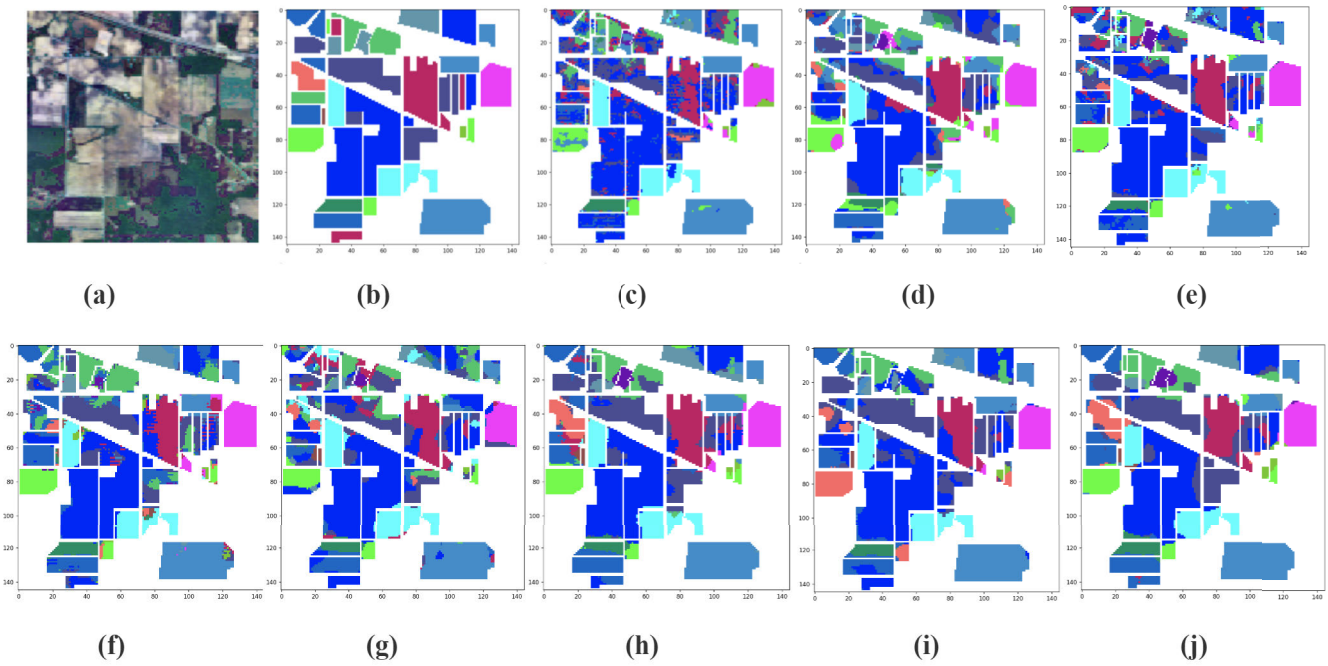


Fig. 12. Classification maps results on IP. (a) Pseudo-RGB image. (b) Ground truth. (c) 2-D CNN. (d) HybridSN. (e) 3-D–2-D CNN. (f) LSSAN. (g) F3D. (h) MSDAN. (i) LiteDepthwiseNet. (j) Lite-HCNet (1% samples for training).

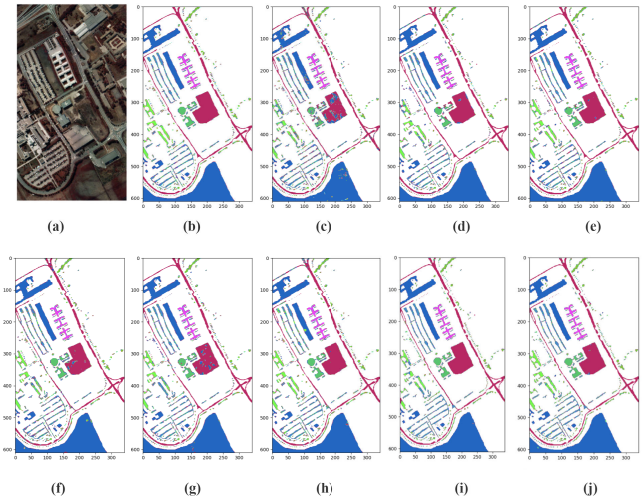


Fig. 13. Classification maps results on UP. (a) Pseudo-RGB image. (b) Ground truth. (c) 2-D CNN. (d) HybridSN. (e) 3-D–2-D CNN. (f) LSSAN. (g) F3D. (h) MSDAN. (i) LiteDepthwiseNet. (j) Lite-HCNet (1% samples for training).

in the IP dataset, some samples had an extracted number of 0 at 1% scale. Therefore, we specifically set the number of samples with an extracted number of 0 in the IP dataset to be uniformly set to 1 at 1% scale.

It is clear from Table II that the proposed Lite-HCNet model achieves the highest classification accuracy on all four experimental datasets with 1% training samples, except for the k metric on the IP dataset. Compared with the classical hybrid convolutional models HybirdSN and 3-D–2-D CNN, Lite-HCNet demonstrates significant accuracy improvement. In particular, the AA of Lite-HCNet is improved by 20% on the IP dataset with highly unbalanced sample distribution. Lite-HCNet also shows considerable accuracy improvement compared with LSSAN, F3D, and LiteDepthwiseNet, which are mainly lightweight and fast. Furthermore, even when compared with the complex MSDAN, Lite-HCNet achieves higher performance, although the k metric on the IP dataset is slightly lower than that of MSDAN by only 0.08%. These results strongly demonstrate that our proposed Lite-HCNet has superior performance under imbalance and small sample conditions.

In addition to the most intuitive classification accuracy results, model complexity and computational cost are also key indicators of a model's performance, directly related to the computational resources required and the ease of deployment. Table III provides a detailed overview of the model complexity and runtime of each model on different experimental datasets with 1% training samples. It is evident from Table III that Lite-HCNet has a significant advantage in terms of the number of model parameters and floating point operations (FLOPs) metrics compared with other models. HybirdSN, for instance, has 1707 times more parameters and 1815 times more FLoPs metrics than Lite-HCNet on the IP dataset. Even the most volume-competitive lightweight model LiteDepthwiseNet has 17 times the number of parameters and 44 times the FLoPs metric of Lite-HCNet. Although the 2-D CNN model achieves the lowest FLoPs metric on the IP dataset, it has almost 100 times the number of parameters and correspondingly much lower classification accuracy than our Lite-HCNet. Furthermore, Lite-HCNet shows similar advantages on all other three datasets. These results convincingly prove that our proposed Lite-HCNet not only has excellent classification accuracy performance with a small number of training samples, but also has superior performance in terms of model complexity and efficiency.

Figs. 12–15 illustrate the comparison of the visual classification results of different methods on four experimental datasets under 1% training samples, respectively. It is evident from the figures that the proposed Lite-HCNet generates higher quality classification maps than other methods. Even with a small number of training samples on the IP dataset, Lite-HCNet produces classification maps with relatively more

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TABLE IV
CLASSIFICATION ACCURACIES (IN PERCENTAGES) ON IP, UP, HU, AND YRD (10% SAMPLES FOR TRAINING)

Dataset		2D CNN	HybridSN	3D-2D CNN	LSSAN	F3D	MSDAN	LiteDepthwiseNet	Lite-HCNet
Indian Pines	$k \times 100$	81.67±0.2	96.99±0.3	97.15±0.5	97.20±0.2	97.70±0.4	**98.44±0.1**	96.69±0.4	98.30±0.1
	OA × 100	83.89±0.2	97.67±0.5	98.01±0.3	97.43±0.2	97.53±0.2	**98.57±0.1**	97.19±0.4	98.50±0.1
	AA × 100	81.73±0.3	95.50±0.4	96.82±0.4	92.28±0.3	96.39±0.5	98.27±0.1	96.12±0.5	**98.46±0.1**
University of Pavia	$k \times 100$	96.84±0.2	99.53±0.1	**99.75±0.1**	99.33±0.1	99.57±0.1	99.43±0.1	99.24±0.1	99.35±0.1
	OA × 100	97.32±0.2	99.59±0.1	**99.74±0.1**	99.51±0.1	99.62±0.1	99.49±0.1	99.47±0.1	99.51±0.1
	AA × 100	96.91±0.2	98.6±0.7	**99.58±0.1**	99.16±0.1	99.48±0.1	99.13±0.1	99.08±0.1	99.18±0.1
Houston 2013	$k \times 100$	96.95±0.3	98.69±0.1	99.03±0.2	98.37±0.2	98.28±0.2	**99.12±0.1**	97.48±0.6	99.00±0.1
	OA × 100	97.24±0.1	**99.73±0.2**	98.76±0.2	98.39±0.1	98.50±0.1	99.24±0.1	97.39±0.7	99.04±0.1
	AA × 100	96.50±0.2	99.02±0.1	98.92±0.2	98.53±0.1	98.15±0.3	**99.31±0.1**	97.56±0.6	99.14±0.1
Yellow River Delta	$k \times 100$	74.72±0.3	89.11±0.4	84.57±0.3	86.81±0.3	**89.33±0.3**	89.15±0.1	88.04±0.2	86.81±0.3
	OA × 100	81.73±0.3	**92.06±0.3**	89.05±0.5	89.95±0.6	91.64±0.3	91.61±0.2	89.96±0.5	89.92±0.2
	AA × 100	73.88±0.2	**90.29±0.2**	85.41±0.3	87.66±0.5	89.82±0.1	89.54±0.3	89.25±0.5	87.13±0.4

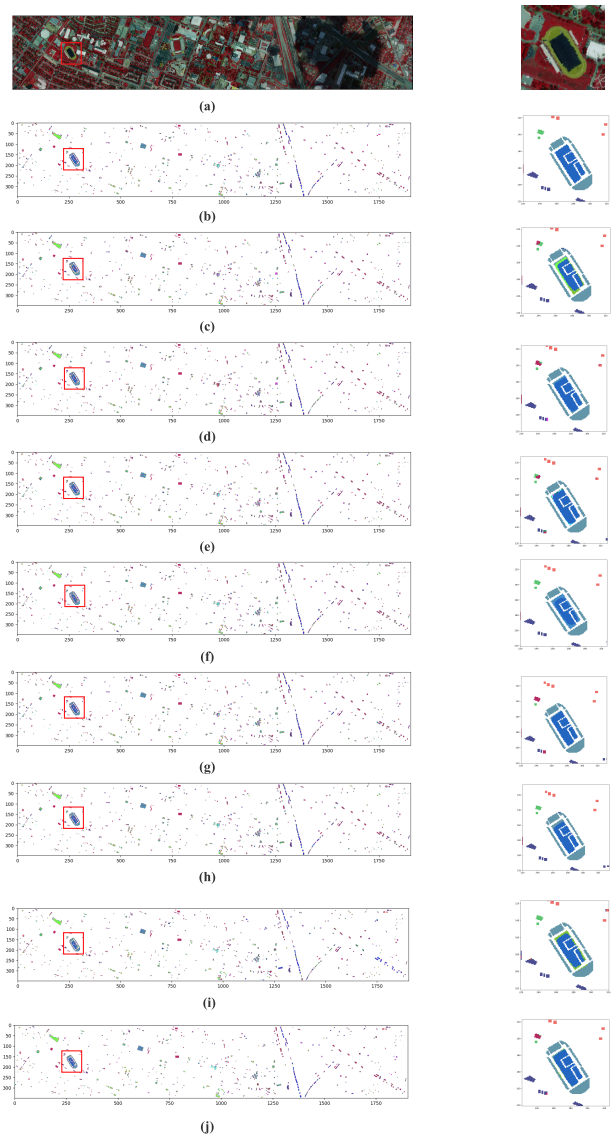


Fig. 14. Classification maps results on HU. (a) Pseudo-RGB image. (b) Ground truth. (c) 2-D CNN. (d) HybridSN. (e) 3-D–2-D CNN. (f) LSSAN. (g) F3D. (h) MSDAN. (i) LiteDepthwiseNet. (j) Lite-HCNet (1% samples for training).

accurate distribution and better edge processing. On the UP and YRD datasets, Lite-HCNet obtains classification maps with smoother distributions and less noise. Even on the more complex HU dataset, Lite-HCNet still produces higher quality classification maps, as can be clearly seen from the enlarged details in the red box.

To avoid the possible overfitting of the model due to a limited number of training samples, we conducted a comparison experiment with 10% of the training samples on four experimental datasets. The experimental results are shown in Table IV, where we observe that increasing the proportion of training samples to 10% leads to improved classification accuracy for all models. Although the Lite-HCNet does not achieve the highest precision due to its lightweight, it still achieves competitive results, with an average difference from the highest accuracy about 0.3% on the IP, UP, and HU datasets. Even the AA metric achieves the highest accuracy on the IP dataset. On the YRD dataset, the average difference from the highest accuracy is about 2.6%. These results show that the proposed Lite-HCNet remains at the forefront, even under adequate sample conditions.

D. Ablation Experiment

To further investigate the effectiveness of the designed LE-DSC, we conducted a series of ablation experiments. Specifically, we replace all LE-DSCs in the Lite-HCNet model with deeply separable convolutions of the same input–output size, as well as typical convolutions, named Lite-HCNet-DSC and Lite-HCNet-NC, respectively, to verify the performance difference of LE-DSC compared with DSC and NC. The experimental results with 1% and 10% training samples are shown in Table V.

As can be seen from Table V, the DSC has a significant decrease in the number of parameters and FLoPs metrics compared with the NC with the same structure, but the accuracy is significantly lower than that of NC. In contrast, the Lite-HCNet constructed based on our designed LE-DSC has a lower volume than the Lite-HCNet-DSC while achieving a higher classification accuracy than the Lite-HCNet-NC. This strongly demonstrates the advanced performance of our designed LE-DSC.

It is also noteworthy that the models constructed using DSC and NC instead of LE-DSC exhibit substantial performance due to the excellent design of our LE-HCL framework.

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TABLE V
EXTERNAL ABLATION EXPERIMENT ON IP, UP, HU, AND YRD (1% AND 10% SAMPLES FOR TRAINING)

Dataset		Lite-HCNet-DSC(1%)	Lite-HCNet-NC(1%)	Lite-HCNet(1%)	Lite-HCNet-DSC(10%)	Lite-HCNet-NC(10%)	Lite-HCNet(10%)
Indian Pines	$k \times 100$	79.11	**80.82**	80.54	97.21	98.01	**98.30**
	OA × 100	81.59	82.42	**82.72**	97.55	98.25	**98.50**
	AA × 100	80.50	82.03	**83.43**	97.74	97.64	**98.46**
	Params/K	4.8	41	**3.1**	4.8	41	**3.1**
	FLoPs/M	116	649	**70**	116	649	**70**
University of Pavia	$k \times 100$	95.46	96.57	**96.86**	99.12	99.30	**99.35**
	OA × 100	96.59	97.42	**97.65**	99.33	99.47	**99.51**
	AA × 100	95.43	96.07	**96.66**	98.98	98.98	**99.18**
	Params/K	3.6	26.3	**2.2**	3.6	26.3	**2.2**
	FLoPs/M	12	60	**7**	12	60	**7**
Houston 2013	$k \times 100$	87.87	89.16	**90.02**	96.71	98.06	**99.00**
	OA × 100	88.81	89.98	**90.51**	96.95	98.21	**99.04**
	AA × 100	88.49	89.62	**90.31**	97.21	98.38	**99.14**
	Params/K	5.5	51.7	**3.6**	5.5	51.7	**3.6**
	FLoPs/M	18	109	**11**	18	109	**11**
Yellow River Delta	$k \times 100$	80.81	81.04	**81.70**	83.52	84.01	**86.81**
	OA × 100	85.38	85.61	**86.06**	87.45	87.94	**89.92**
	AA × 100	81.76	81.69	**82.24**	84.71	84.09	**87.13**
	Params/K	5.4	51.6	**3.5**	5.4	51.6	**3.5**
	FLoPs/M	45	269	**27**	45	269	**27**

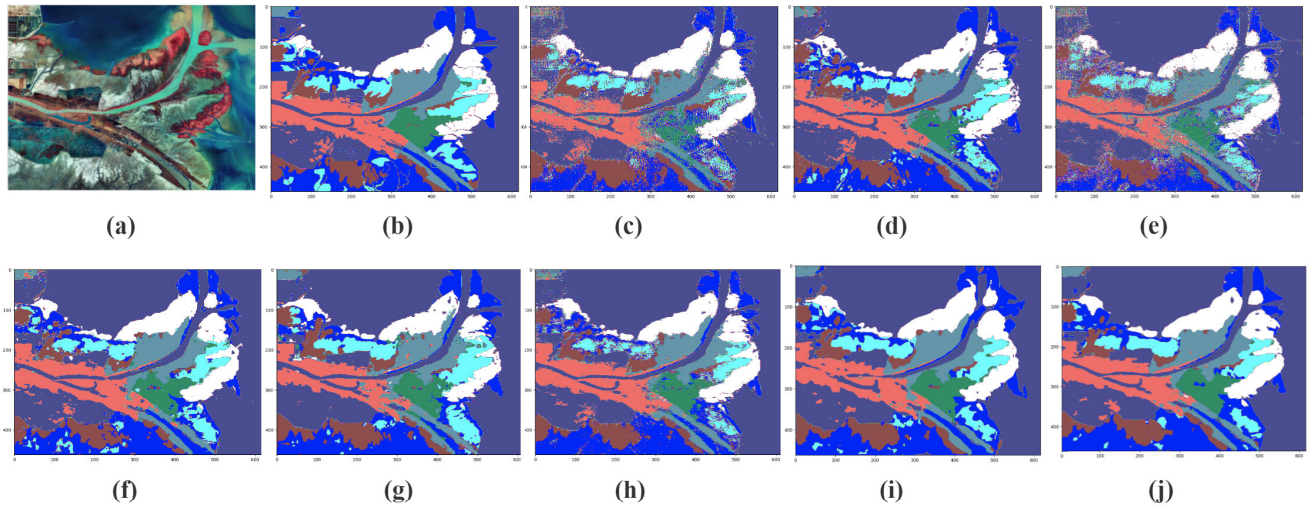


Fig. 15. Classification maps results on YRD. (a) Pseudo-RGB image. (b) Ground truth. (c) 2-D CNN. (d) HybridSN. (e) 3-D–2-D CNN. (f) LSSAN. (g) F3D. (h) MSDAN. (i) LiteDepthwiseNet. (j) Lite-HCNet (1% samples for training).

To further explore the specific contribution distribution of the internal structure of LE-DSC, we designed more detailed ablation experiments. Specifically, we used Lite-HCNet-DSC as the baseline model and performed experiments by adding our designed modules or modifying the corresponding structures to all the DSCs in it. Among them, Lite-HCNet-DSC-CKD refers to changing the DSCs in Lite-HCNet-DSC to the CKD structure. Lite-HCNet-DSC-NCAM refers to adding NCAM to the DSCs in Lite-HCNet-DSC. Lite-HCNet-DSC-SE and Lite-HCNet-DSC-ECA refer to adding SE and ECA to the DSCs in Lite-HCNet-DSC, respectively. The experimental results under 1% and 10% training samples for the UP dataset are shown in Table VI. We noticed that ECA has a lower number of parameters and FLoPs than SE but lower accuracy, while our NCAM has a higher accuracy and a more balanced number of parameters and FLoPs. Meanwhile, we observed an interesting issue that the CKD operation reduces the model's volume, resulting in higher gains under small training samples but poor results under large training samples. NCAM has a small volume increase compared with DSC, resulting in poor performance under small training samples but higher gains under large training samples. The combined LE-DSC has a more robust structure, with a lighter volume and excellent performance under both small and large training samples.

V. CONCLUSION

In this article, we optimized the traditional DSC and proposed the LE-DSC, which combines the CKD strategy with the designed NCAM. Compared the traditional DSC, the proposed LE-DSC has fewer parameters and requires less computation, while still achieving more advanced feature extraction capability. The LE-HCL built based on LE-DSC further explores efficient connection method of hybrid convolution and maximizes the advantages of feature extraction. In addition, we developed a lightweight neural network, the Lite-HCNet, which uses the LE-HCL as its core feature extraction network for HSI classification. Despite containing only a small number of parameters and requiring minimal

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TABLE VI
INTERNAL ABLATION EXPERIMENT ON UP (1% AND 10% SAMPLES FOR TRAINING)

	Model	$k \times 100$	OA	AA	Params/K	FLoPs/M
1%	Lite-HCNet-DSC	95.46	96.59	95.43	3.6	12.3
	Lite-HCNet-DSC-CKD	96.10	97.05	96.01	**2.1**	**6.6**
	Lite-HCNet-DSC-NCAM	95.75	96.81	95.54	3.7	13.1
	Lite-HCNet-DSC-SE	95.29	96.46	94.46	6.4	12.7
	Lite-HCNet-DSC-ECA	93.67	95.24	93.54	3.6	12.6
	Lite-HCNet	**96.86**	**97.65**	**96.66**	2.2	7.4
10%	Lite-HCNet-DSC	99.12	99.33	98.98	3.6	12.3
	Lite-HCNet-DSC-CKD	98.64	98.97	98.32	**2.1**	**6.6**
	Lite-HCNet-DSC-NCAM	99.31	99.48	99.15	3.7	13.1
	Lite-HCNet-DSC-SE	98.96	99.22	98.84	6.4	12.7
	Lite-HCNet-DSC-ECA	98.95	99.21	98.84	3.6	12.6
	Lite-HCNet	**99.35**	**99.51**	**99.18**	2.2	7.4

computational effort, extensive experiments demonstrated that the Lite-HCNet outperforms other methods, especially when dealing with limited training samples. Moreover, our sufficient ablation experiments confirmed the effectiveness of the designed LE-DSC.

The Lite-HCNet is a simple model that achieves satisfactory performance without in-depth refinement. Moving forward, we plan to optimize the LE-DSC and even further develop more effective network models based on its architecture.

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